Haemoglobin Estimation by Sahli's Method

Learning Objectives

- By the end of this session students should know about:-
- Basic structure of hemoglobin
- Functions of Hemoglobin
- Various laboratory methods for estimation of hemoglobin
- Enumerate the advantages and disadvantages of sahli's method.

Structure of haemoglobin

- Hemoglobin is composed of four polypeptide chain and four heme groups.
- α (Alpha) chain consists of 141 Amino Acids
- β (Beta) chain has 146 amino acids.
- Molecular weight is 64,458-Dalton
- About 6025 grams of hemoglobin is synthesized each day to replace the hemoglobin lost due normal destruction of erythrocytes
- Symbol Hb.

- Synthesis begins from Proerythroblast to Reticulocytes in the bone marrow.

- Hemoglobin A is normal adult hemoglobin and hemoglobin F is fetal hemoglobin.

- The absence, replacement, or addition of only one amino acid modifies the properties of the hemoglobin.

NORMAL VALUES OF HEMOGLOBIN

- Adult Male: 14-16 gm/dl
- Adult Female: 13-15 gm/dl
- Newborn: 16-18 gm/dl

CLINICAL SIGNIFICANCE OF HEMOGLOBIN ESTIMATION

- Hemoglobin estimation gives a brief idea of the pathological conditions to the physician so that your physician can easily understand the cause of pathology and prescribe an effective treatment for it.

Hemoglobin is increased in following conditions

- Polycythemia Vera
- Congenital Heart disease
- High Altitudes
- Heavy smoking
- Methemoglobinemia
- Elevated erythropoietin levels
  - Tumors of Kidney, Liver, CNS, Ovary etc.
  - Renal Diseases (Hydronephrosis& Vascular impairment)
- Adrenal hypercorticisim
- Relative causes of high hemoglobin content
  - Dehydration – Water deprivation, Vomiting, Diarrhea
  - Plasma loss – Burns,

Haemoglobin decreases in

- Iron deficiency anemia
- Parasitic infections severely in hookworm infection
- Sickle cell anemia
- Thalassemia
- Aplastic anemia
- Hemolytic anemia
- Loss of blood
- Leukemia
- Tuberculosis

FACTS ABOUT HEMOGLOBIN

- Each gram of hemoglobin carries 1.34 ml oxygen
- Each gram of Hb contains 3.33 mg of iron
- Total Hb content in a healthy body is approximately 600 gms.

Functions of haemoglobin

- The main function is to deliver Oxygen to the tissue and carbon dioxide from tissue to the lungs.

- It imparts red color to the blood.

- It buffers blood pH.

Indications of estimating haemoglobin

1. To determine presence and severity of anemia.
2. To detect the oxygen carrying capacity of blood.
3. Estimation of Red cell indices.
4. To check haemoglobin level of blood prior to donating blood.

Normal Values

The normal value depends on the age and sex of the individuals:

- Men: – 13-18 gm/dl
- Women :– 11-16 gm/dl
- Full term / cord blood :– 13-19 gm/dl
- Children 1 year :- 11-13 gm/dl
- Children 10 - 12 years :- 12-15gm/dl

Different methods of Estimation of Hemoglobin

- Sahli's Method
- Colorimetric Method
- Specific Gravity method
- Gasometric method
- Chemical Method
- Automatic count technique

Principle

- When blood is mixed with an acid solution, the hemoglobin converts into the brown-colored acid hematin. The acid hematin is then diluted with distilled water till the color of the acid hematin matches that of the brown glass standard. The hemoglobin is estimated by reading the value directly from the scale.

Equipment

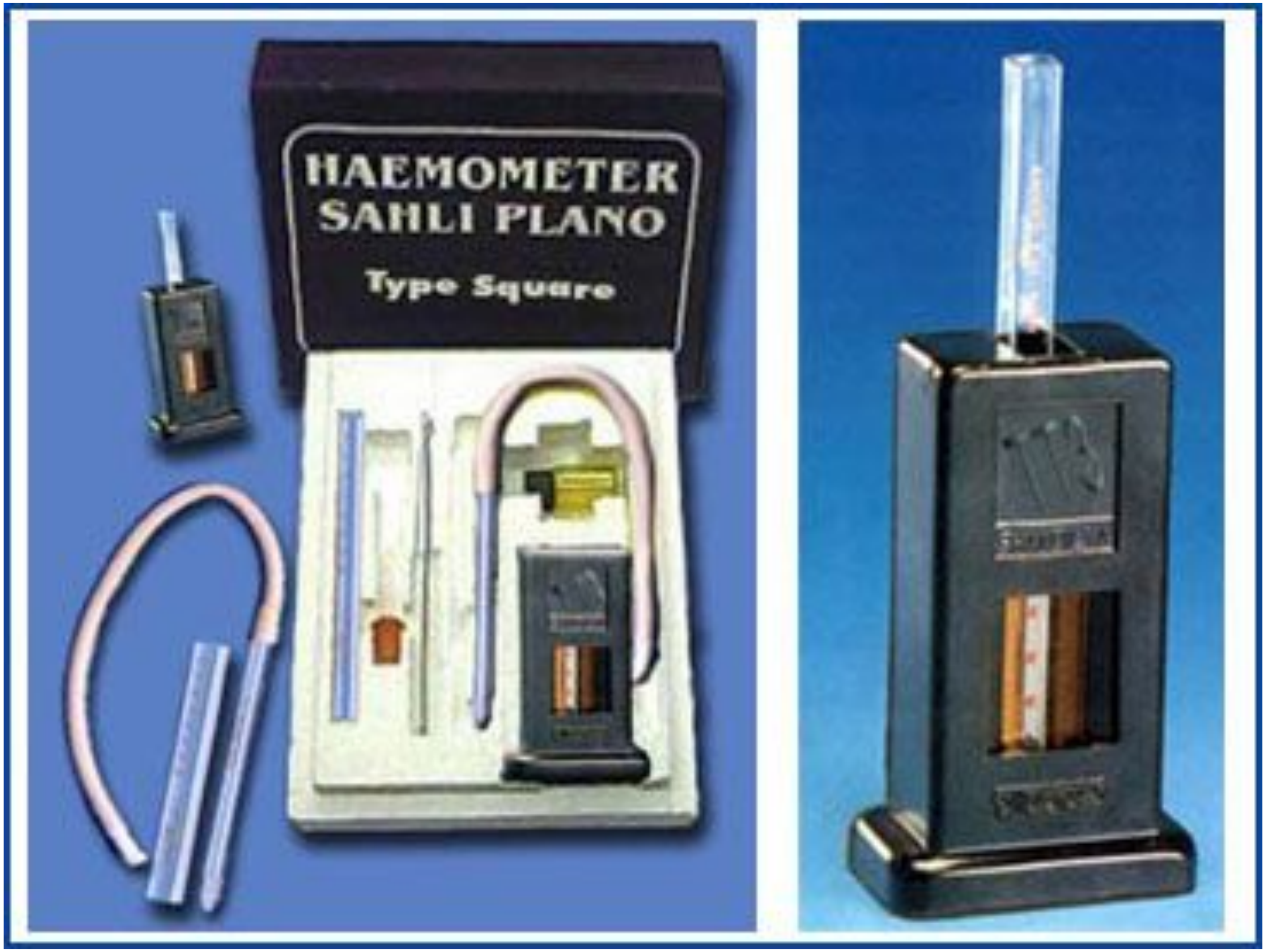
- Sahli's hemoglobinometer: This equipment consists of a comparator with a brown glass standard and Sahli's graduated hemoglobin tube which is marked in percent and gram.
- Hemoglobin pipette or Sahli's pipette (marked at 0.02 ml or 20 μl).
- Stirrer (a small glass rod).
- Dropping pipette (dropper).

Reagents

- Distilled water
- N/10 Hydrochloric acid (HCl) ,0.1 N HCl Prepared by diluting 0.8 ml conc. HCl with 99.2 ml D.W.

Specimen

- Blood is obtained directly by skin puncture or EDTA-anti-coagulated venous blood.


HAEMOMETER
SAHLI PLANO
Type Square

Method

1-The N/10 hydrochloric acid is placed into Sahli's graduated tube up to mark 2 grams.

2-Sterilize the fingertip with alcohol swab and allow it to dry

3-Using sterile lancet prick the finger tip and wipe away first drop of blood

4-Suck blood into the hemoglobin pipette (capillary pipette) up to 20 cu.mm (up to 20 mm mark) avoid air bubbles coming into a tube .

5- Blood adhering to the outer part of the pipette is wiped away with the help of tissue paper (absorbent paper) or cotton (gauze piece).

6-Add blood sample to the N/10 hydrochloric acid solution which is placed into Sahli's graduated tube, mix the mixture with the help of a glass stirrer, and allow the tube to stand for 10 minutes.

Method----Continued

- Add distilled water drop by drop into the mixture placed in Sahli's graduated tube, till the color of the solution matches that of the brown glass standard.
- Take the reading of the lower meniscus from the Sahli's graduated tube in grams.

Advantages:-

- No time required for colour development
- This is simple and accurate method
- Economical and easy to perform
- It is fast and accurate method than visual comparative method
- Disadvantages:-
- Standard solutions not easily available and unstable
- Methahemoglobinemia and carboxyhemaglobin are not accurately detected.

Q-1-What is hemoglobin

Hemoglobin is an iron-rich protein in red blood cells that carries oxygen from the lungs to the rest of the body. It's crucial for delivering oxygen to cells, allowing them to function properly

Q-2-What are the functions of Hemoglobin?

Hemoglobin's primary function is to transport oxygen from the lungs to the body's tissues and to carry carbon dioxide from the tissues back to the lungs for exhalation. It also plays a role in regulating blood pH by acting as a buffer

Q-3-What is the normal concentration of hemoglobin in males and females?

Normal hemoglobin levels differ between males and females. In adult males, the typical range is 13.5 to 17.5 grams per deciliter (g/dL), while in adult females, it's 12.0 to 15.5 g/dL according to MedlinePlus and the Mayo Clinic. These values can vary slightly between different medical sources and laboratories, according to the Mayo Clinic.

Q-4-How much O2 is carried by one gram. Of hemoglobin?

One gram of hemoglobin can carry 1.34 milliliters of oxygen when fully saturated. This is the maximum amount of oxygen that a gram of hemoglobin can bind.

Q-5-What is the function of iron in the hemoglobin?

Iron's primary function in hemoglobin is to bind and transport oxygen. Specifically, each iron atom within the hemoglobin molecule can bind to one oxygen molecule, and since hemoglobin contains four iron atoms, it can carry four oxygen molecules. This process is crucial for delivering oxygen from the lungs to the body's tissues.

Q-6- What are the normal types of hemoglobin?

Hemoglobin (Hgb) A, the most common type of hemoglobin in healthy adults.

Hemoglobin (Hgb) F, fetal hemoglobin. This type of hemoglobin is found in unborn babies and newborns. HgbF is usually replaced by HgbA by age one to two years.

Q-7-What is anemia?

Anemia is a condition where the body doesn't have enough healthy red blood cells to carry adequate oxygen to the body's tissues. This can lead to various symptoms like fatigue, weakness, shortness of breath, and dizziness. Anemia can be caused by a variety of factors, including iron deficiency, vitamin deficiencies, chronic diseases, and inherited conditions.

Q-8-Name the types of Anemia

Anemia, a condition characterized by a deficiency in red blood cells or hemoglobin, has several types. These include iron-deficiency anemia, vitamin-deficiency anemia (including B12 and folate deficiencies), aplastic anemia, anemia of chronic disease, hemolytic anemia, sickle cell anemia, and thalassemia

Q-9-What is color Index?

The color index, taken as the ratio of the percentage of hemoglobin (compared to "normal") to the red blood cell count (compared to "normal") has been a useful clinical figure in differentiating iron deficiency from other types of anemia.

Q-10-are the various methods of measuring Hb concentration?

The Haemospect® device uses reflection spectroscopy.

Manual laboratory-based methods. The cyanmethemoglobin method (Drabkin's method) was established in the 1950s and is accepted as a reliable method of Hb measurement. ...

Automated hematology analyzers. ...

Hemoglobin meters. ...

Hemoglobin color scale. ...

Noninvasive methods.

Q-11-Why is N/10 HCl used in estimating Hb by Sahli's method?

In Sahli's method, N/10 HCl is used because it converts hemoglobin to acid hematin, a brown-colored compound, and the intensity of this color is directly proportional to the hemoglobin concentration. This specific concentration (N/10) is crucial because the method is standardized for it, and using a different concentration could damage the hemoglobin proteins and interfere with the accuracy of the measurement

Q-12-What is the importance of acid hematin formation?

The acid hematin method, also known as Sahli's method, is a technique for estimating hemoglobin concentration in blood by converting hemoglobin to acid hematin and comparing the resulting color to a standard. The procedure involves mixing blood with N/10 hydrochloric acid, allowing the hemoglobin to convert to brown acid hematin, and then diluting the solution with distilled water until its color matches a standard. The hemoglobin concentration is then read directly from a graduated tube